



flows through it. The device, commonly called a solenoid, includes a coil made of a long wire and a moving core (plunger) made of iron or another magnetic material. Applying current to the coil moves the core, enabling mechanical actuation.

A voltage is applied to create a magnetic field when actuating a solenoid. Due to the coil's inductance, current build-up takes time. Since the force on the solenoid's core is proportional to the current, sufficient voltage must be applied to quickly establish the current and generate maximum force to move the core. The DC input voltage for



Fig. 4: A12V DC push-pull solenoid

this solenoid pulse driver circuit is 12V as it is designed for a 12V DC solenoid.

The protection fuse F1 should be selected based on the solenoid's current rating to prevent unnecessary blowing. Additionally, the fuse's voltage rating must equal or exceed the circuit's voltage. During testing, a 600mA slow-blow cartridge fuse was used, as the 12V DC push-pull solenoid in the prototype has a 300mA current rating (see Fig. 4).

The operation is straightforward. Initially, the monostable (IC2C and IC2D) output remains off in the standby state. The 10kΩ resistor (R2) is a pull-up resistor, keeping the IC2A input high unless a valid input signal is received through the PC817 optocoupler (IC3).

When an input signal is received via IC3, the 10μF capacitor (C7), which is uncharged in the standby state, starts charging, initiating the time period. The TTL input trigger signal has no further effect until the output returns to its normal, stable state. Capacitor C7 and resistor R3 determine the time period; for most logic gates, the theoretical timeout period is given by: $T = 0.7 \times R3 \times C7$

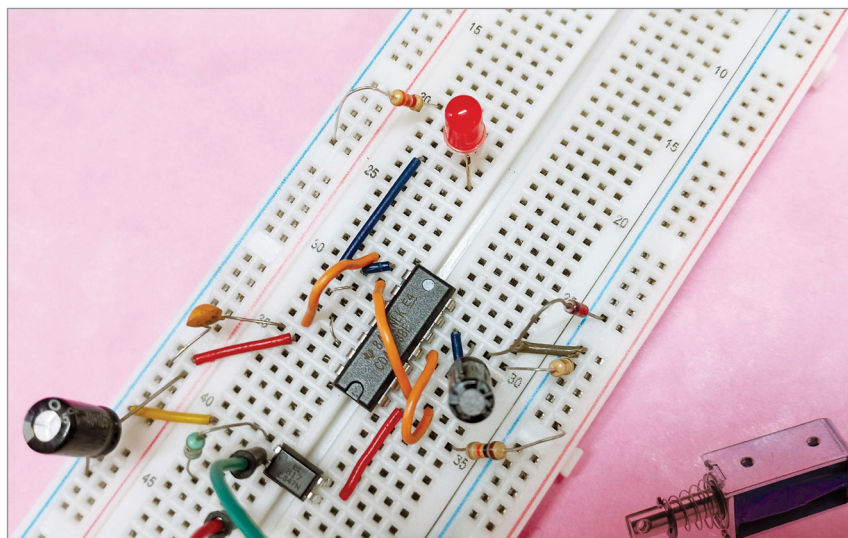


Fig. 5: Author's prototype

PARTS LIST

Semiconductors:

IC1	- LM7805 voltage regulator
IC2 (ICA-IC2D)	- CD4093 Quad NAND gate
IC3	- PC817 optocoupler
T1	- IRLZ44 MOSFET
D1-D3	- 1N4007 rectifier diode
D4	- 1N4148 signal diode
LED1	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1	- 330Ω
R2, R5	- 10kΩ
R3	- 150kΩ
R4	- 100Ω
R6	- 3.3kΩ

Capacitors:

C1	- 1000μF, 35V electrolytic
C2, C5, C7	- 10μF, 35V electrolytic
C3, C4, C6	- 100nF ceramic disc

Miscellaneous:

CON1	- 2-pin connector
JACK1	- DC jack
F1	- 600mA slow-blow cartridge fuse
S1	- Push-to-on switch
	- 12V DC adaptor

In brief, the solenoid activates when it receives a pulse through IC3 and deactivates when it does not.

The optional debug indicator LED1 represents the monostable's output state. Likewise, the push-to-on switch S1 is an optional manual test button.

Construction hints

A breadboard assembly is sufficient for quick testing, but for regular use, constructing the circuit on a perf-board or a custom-printed circuit board is recommended. Fig. 5 shows the author's preliminary breadboard setup.

This circuit demonstrates that a pulse driver for standard solenoids can be implemented at a low cost using readily available components. Solenoid drivers are widely used for linear and rotational actuation in mechanical systems. **EFY**



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